

Full English guide (version 5.1)

<https://youtu.be/YlxmJX4KJm8>

Addition to the guide (version 6.0)

- Added Sword animation polishing tool

<https://youtu.be/PgtqwtHpjiQ>

A set of tools for polishing animations. Both made by hand and obtained with the help of a mocap

Multiplies and simplifies the process of animating characters and their interactions with each other and the world.

Helps you solve complex technical issues that arise during animation.

If your Rig does not have any functions needed for this particular animation, you can add them without involving the Rigger.

It allows you to get rid of the constant switching to different spaces in the rig, to get rid of the IK-FK switch.

And much more.

General approach to animation using scripts:

Sketch the animation - the main focus is on good animation dynamics, skipping moments with interactions or marking them roughly. After that, you can start polishing with scripts.

[basic scripts for the OverRig animation method](#)

knots

KNOT

A locator is created for the selected objects, the animation of the object in world coordinates is transferred to it, and the object is postponed to the corresponding locator. You can do it on many objects at once.

If you apply this script with a dedicated timeline, the created locators will have an "attach" channel, in which the weight will highlight the working interval. By animating this channel, you can control the locator's effect on the object.

Using:

- The most used script. A basic locator for independent work or subsequent more complex interaction.
- You need to fix the shaking of a control that is inside other controls? Hanging it on KNOT - and you fix the animation.
- Need to not slip feet in the gait animation, which is made on the spot?
- Moved it to KNOT - and tweaked it.
- [Walking. Correcting sliding. Convert to progressive cycle. Freeze main.](#)
- Do you want to reset the MAIN CONTROL, or move it to a more convenient location?
- Moved the next objects in the hierarchy to KNOT and moved them.
- And much, much more.
- [Flip on horizontal bar. OverRig animation method](#)

KNOT OFFSET

A locator is created for the selected object, which must be moved and, if necessary, rotated to the desired location. After that it is necessary to click in an empty place on the screen (unselect everything). The animation of the object relative to this point is transferred to the locator and the object will be postponed to it. The object will move after the locator, but now will have a different pivot.

You can do it on a lot of objects at once

If you apply this script with a dedicated timeline - then the created locators will have an "attach" channel in which the weight will highlight the working gap. By animating this channel, you can control the locator's effect on the object.

Using:

- Want to rotate the weapon around another axis? Assign it - and move the locator to the desired point.
- [Toss up Kettlebell. Object manipulation. OverRig animation method](#)

- Do you want the animation to have no gimbal locus at the right part of the animation? Assign it, rotate it on the right axis - and just fix the animation as needed.
- [Interaction with enviroment. OverRig animation method](#)
- Do you want the foot to rotate around the toe and not slip in the mocap? - Move the foot control to this point and remove the keys to move it!
- And much, much more.

KNOT ORIENT

A locator is created for the object, which controls the transferred animation of the object's rotation in world coordinates, but the locator itself follows the object.

You can do it on many objects at once

If you apply this script with a dedicated timeline, the created locators will have an "attach" channel where the working interval will be highlighted by weight. By animating this channel, you can control the locator's effect on the object.

Using:

- Assigned to the fingers - you twist the hand, but the fingers do not twist with the hand.
You want to rotate your torso so that your head doesn't rotate - hang it on your head and rotate your torso.
- If you want to animate the tail, but make it independent of the body, put it on this locator.
- And so on..

Smart object

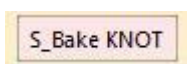


A set of basic nodes, but which bake the animation according to the frames of the original object

Added by request of traditional animators

Using:

- Switching spaces on the controls of the Rig (I sleeved on the SmartKNOT, switched the space on the control, and sleeved again frame by frame with S_Bake KNOT)
- Work with the rig at the stage of sketching, if you do not want the objects to be baked in each frame



Bakes the animation in the same keys as on the original node

parent

A powerful set of scripts for interacting with objects and with other characters

INSIDE

The selected objects will move inside the last selected object with the rotation and movement animation saved. In order to avoid rig breakage - controllers must be moved to KNOT beforehand

Using:

An indispensable script for interacting with objects.

- Hang the apple on KNOT, then highlight the hand control and press Parent Inside. The apple is now inside the hand. You can now animate its movement relative to the hand. If you want to put it in the other hand, just highlight the locator, highlight the other hand and Parent Inside
- Do you want your character to grab the enemy by the throat with his hand and not let go? - Control your hand to KNOT, then highlight your enemy's throat control and Parent Inside. After that, set the grab pose and delete all unnecessary keys!

WORLD

All selected objects are transferred to the world coordinates with saving the animation of movement and rotation. In order to avoid rig breakage - it is necessary to transfer controls to KNOT beforehand

Using:

- Is it time to drop the apple and the previous example you just twirled in your fingers? Highlight the apple locator and click this script. Now you can animate the whole thing in world coordinates.

SWAP

The selected objects will be placed inside the last selected object, preserving the animation. It also puts these objects inside the hierarchy where the first selected object was.

It differs from the INSIDE script in that it can swap nested objects.

Controls must be on KNOT beforehand.

Using

- Sometimes it is better that not the apple listens to the hand, but the hand listens to the apple. Select the wrist locator first, then the apple locator, and click. They will swap places in the hierarchy, but the apple will be in the same hierarchy as the wrist.

Across

Selected objects will be baked onto KNOT and placed inside the last selected .
This is a quick substitute for the two scripts KNOT and PARENT_INSIDE

Using

- Often, when working with controls, in order not to break the rig, it is necessary to bake the animation of controls inside other objects not directly, but through KNOT. And in order not to do unnecessary operation - use this script. And also it works through the timerange.

ATTACH



ATTACH channel control on KNOT objects. Works also with dedicated key timerange

Using

- The ATTACH channel is a powerful animation control tool. Allows you to assign multiple KNOTs to one object and control the effect on it

aim

AIM

For the selected object, two locators TOP and BOX are created, which you need to move to convenient points. After that it is necessary to click with the mouse in an empty place on the screen (deselect everything). And the animation of rotation of the object is transferred to the locator. To select these objects, use buttons 1 and 2 in the Selector section. Or left and right arrows on the keyboard (standard Maya hotkey)

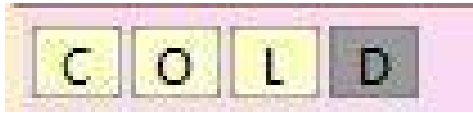
You can do many objects at once

Using:

- Animation of tails, wings, clothes!
- [EazyTail animation with OverRig animation method](#)

- Managing object interaction
- Pitching animation

Key tools



C

Makes pre/post infinity cycle on selected objects

O

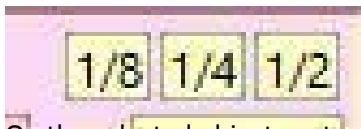
Makes pre/post infinity oscillate on selected objects

L

Makes pre/post infinity linear on selected objects

D

Makes pre/post infinity default on selected objects



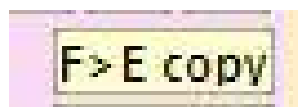
On the selected objects puts a double oscillating key (sinusoid), equal to $1/8$, $1/4$, $1/2$ period of the current timerange.

Using

- Indispensable for making looped animations (running, walking, swimming, etc.) by the sinusoid method



Shift the entire animation by 1, 0.1 frame to the right or left.



Copies the position of the first frame of the timeline to the last frame.

Using

- Making looped animations, when working through animation layers

<1 <0.1

sequentially shift animations of selected objects by 1 or 0.1 frame to the left

0.1> 1>

sequentially shift animations of selected objects by 1 or 0.1 frame to the right

Using:

- Indispensable in combination with the AIM script. Shifts animation of selected objects one position farther than the previous one.
- Works not only with all animation, but also with selected keys.



Places the key on the first and last frame on the timeline

Using

- When working through animation layers - very useful



Places the same key on the selected object within the selected time interval from the first selected frame - for the entire interval (depending on which way the timerange is pulled)

Применение

- Greatly speeds up the work with the mocap
- Indispensable when working with layers
- Allows you to apply interesting methods of working with animation



If no keys are highlighted - puts the mirrored half-period keys according to the

the law of reflection of the animation through the X axis

(for example, in the case of  on the translate X axis - puts a mirror key - leaves the rest unchanged, and on the rotation axis - mirror all but X)

If the keys are highlighted, it mirrors these keys by the law of reflection through the x-axis

Using:

- When creating looped animations (running, walking, standing, etc.) using the sinusoid method
- When refining gait animations derived from the mocap



Copies and pastes the selected curve

If you select only 1 key - it will paste the curve as it was copied.

If you select a curve interval - it will insert in this interval and scale by time

Using

- When creating looped animations (running, walking, standing, etc.) using the sinusoid method
- When refining gait animations derived from the mocap



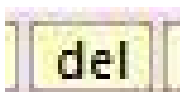
Shifts the animation by half a period (between the start and end of the playback Range)

Using

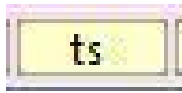
- When creating looped animations (running, walking, standing, etc.) using the sinusoid method
- When refining gait animations derived from the mocap



By the allocated timestring highlights the keys



Deletes selected gaps on all channels on the baked timeline (just select keys on one curve - it deletes all selected gaps) Handy when working with a mocap



By selected keys or timeline puts a toggler to switch to this interval or to the whole timeline. Convenient when working with large animations



Moves the timeslider to the middle of the selected frames

Selector

Src

selects the OBJECT of the last applied script

1

selects the locator of the last applied script (or the TOP locator of the AIM script)

2

selects the second locator of the last applied script, if there is one (or the SIDE locator of the AIM script)



Highlights objects that are attached to the selected KNOT locator

double knot

Local

It creates a system of two nested objects on the selected object - it bakes the animation of the object on the top object, and the object itself is constricted to the inner object

You can do many objects at once

Using:

- Powerful animation editing tool. Especially the animation mocap.
- In essence, by applying it, you rotate and move the object in the axes of the baked animation.
- Allows you to easily intensify movements and make them more expressive by scrolling body parts within an animation already made.
- It gets rid of problems with gimbal lock.

- And much more

World

Creates a system of two nested objects on the selected object - the animation of the object is baked onto the inner object, and the object itself is constricted to the outer object

You can do many objects at once.

Using:

- Allows you to bend the animation in world axes
- [Double World Knot. Bending animation. Improving discobolus animation.](#)

Snapshot Object

Locators are created and animation is transferred from the selected objects to world coordinates.

But the object does not constrain to the locator.

You can do many objects at once.

- Needed for many technical tasks.
- Allows you to animate on the principle of "reverse motion"
- Allows you to take the animation off a complicated slow rig, and attach it to another rig without slowing down the scene even more.
- And much more.

Component

Locators are created and animations are transferred from selected components (vertices, polygons, edges) to world coordinates. But the object does not constrain to the locator.

You can do many objects at once

- Needed for many technical tasks.
- Allows you to animate on the principle of "reverse motion"
- Allows you to remove the animation from a complex slow rig, such as a vertex on a dynamic fabric, and attach it to another rig without slowing down the scene even more.
- And much more.

Particles

Locators are created and animations from selected partials are transferred

And also added a particle speed channel

Hierarchy

Forward

A chain of bones is created from the selected objects in such a way that these bones always face the next object with the X-axis. Inside each bone there is a locator, to which the original object is attached.

Using:

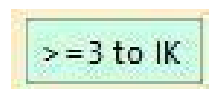
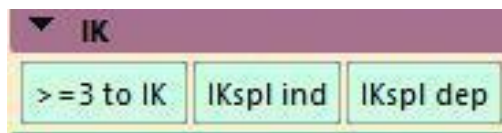
- If you benefit from presenting your character animation as a forward hierarchy.
- [Handstand animation. OverRig animation method. Using Forward Hierarchy script](#)

Reverse

A chain of bones is created over the selected objects, to which the animation is transferred - but in this chain, the animation is transferred from each following object to each preceding bone. And the objects are pre-ordered in the same order.

- Allows you to get an inverse hierarchy on the bones to which the geometry is directly skinned.

IK



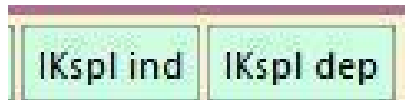
[New Features of Base OverRig scripts](#)

Bakes a forward chain of 3, 4 or more objects on the inverse kinematics system.

Warning. In the frame, in which you run the script, the chain must be at least slightly bent!

Internal objects must not move relative to each other - but can rotate in all directions.

Allows to control any forward kinematics. For example fingers, wings, animal legs, etc. If there are 4 or more objects, internal controls are created to control the shape.



Bakes a forward chain on the IR spline system. Internal objects can both move relative to each other and rotate.

They differ from each other in that the first button creates independent spline controllers, and the second - the internal controllers depend on the movement of the external.

- Allows you to control any forward kinematics. For example, fingers, especially convenient when fingers are straight, tails, torso, etc., etc.

Bake

Bake

Bake animation on selected objects. Works within the minimum and maximum frames of the entire scene

To layer

Bake animation on selected objects into a layer. Works within the minimum and maximum frames of the entire scene.

If the object is under the influence of a node, then to see the difference between what was and what it is now, you must remove the node.

DelCh (НОВОЕ)

removes channels like BlendParent (BlendOrient)etc., which are formed after baking node animations.

It also removes MaxHandles channels - channels that are created when importing skeletons and models from AutodeskMax

Vp

If something went wrong in any script, and the viewport of the program became gray or stopped responding - then click this button!

Sword Tool (НОВОЕ)

This set of scripts is designed for polishing weapon animation.
It is especially relevant for game animators.

Arc Polish

What is he doing?

- Runs the selected object along a smoothed curve with the ability to control its shape and acceleration along it

how it works

- bakes the trajectory of the selected object on a curved trajectory.
- If "show source" ON - an additional curve will be created, showing the original path
- If motion trail is enabled, then motion trail will be enabled within the boundaries of the tool (it slows down the work)
- If besier is enabled, then a besier curve will be created

using

- Specify the number of Knots controls
- select TimeRange or Keys. The movement of the selected object will be transferred to a smooth curve
- button 1 - select the source object
- button 2 - select an object on the curve (to correct the speed and acceleration on the curve)
- button 3 - select the entire arc polishing system (usually to remove after baking)

Make Sword

What it does?

- allows you to adjust the trajectory of the tip of the weapon, which itself rotates and moves in the palm of your hand
- correction will be transmitted to the wrist while maintaining the relative rotation of the weapon,
- the length of the whole arm will not change

how it works

- corrects the rotation of the selected object (weapon and wrist)
- with 3 locators, one of which controls the TOP of the rotation and the other the SIDE

- and an additional system with a special locator that transmits arm movement and the relative movement of the weapon

using

- After starting there will be 2 locators, the first highlighted - TOP - immediately pull it to the top of the weapon.
- After that, either the button in the menu - or the standard hotkey - right arrow
- - select SIDE and pull it in a perpendicular direction to the blade.
- And when everything is ready, click on the empty space in the scene. The script will recalculate

Make Reverse

What it does?

- allows you to lock the weapon at any time (e.g., if it is jammed into the ground)

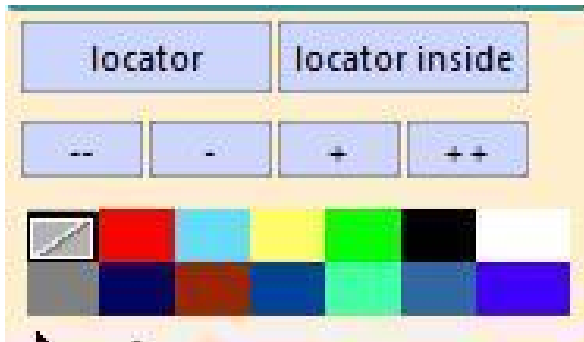
how it works

- bakes the rotation of the selected object (weapon and wrist (and something else))
- using 3 locators, one of which controls the TOP of rotation, the second SIDE
- and PIVOT weapons

using

- After launching, three locators will appear,
- the first highlighted PIVOT - move it to the place where you want to fasten the weapon.
- After that, either the button in the menu - or the standard hotkey - right arrow
- select - TOP - immediately pull it up to the handle of the weapon.
- After that either the button in the menu - or the standard hotkey - right arrow
- - select SIDE and pull it in a perpendicular direction to the blade.
- And when everything is ready, click on the empty space in the scene. The script will recalculate

Miscellaneous



- Auxiliary scripts that allow you to create a normal locator on selected objects (vertexes, etc.)
- Or create a locator inside the selected object
- and change the color and size of the locators
- They also help control the visual size and color of selected locators

Important!!!

Only the direct application of the scripts is described here. But the main power and strength lies in their combination.

- This allows you to get complex combinations of objects that solve the problem in the simplest way.
- it either saves a lot of time, or allows in principle to make such animations, which are not solved by hands in a reasonable time.
- Also all these scripts are used in special effects animation, camera animation, etc..
- Everywhere there is animation - there is a field of application for these scripts.
- The more you use them, the more you find your own ways.
- Beware! They are addictive. Then it is very difficult to do without them.!!!!